PROJECT REPORT

MedCost: honest estimation of medical insurance charges

REDOANUZZAMAN · 29 September 2026

ABSTRACT

A popular open-source *Medical Insurance Cost Predictor* reports $R^2 = 0.953$ for a Random Forest trained on the public US insurance dataset. We show that this figure is inflated by data leakage: the published CSV contains 1,435 exact duplicate rows out of 2,772, so 80% of its test rows also appear in training. Re-evaluated on unique, unseen people the same model scores $R^2 = 0.884$ and a mean absolute error (MAE) of $2,458. We rebuild the pipeline with leakage-free evaluation, a smoker × BMI interaction feature and a Huber-loss gradient-boosting model, cutting honest MAE to $1,154 ($R^2 = 0.928$) while dropping sex and region from pricing. Each estimate comes with a 90% prediction range from group-wise split conformal calibration, which covered 95% of held-out people. The model ships as a web application with what-if scenarios and out-of-range warnings.

1. Introduction

Health-insurance charges depend strongly on a few personal factors, which makes the problem a standard teaching example for regression. For the person using an estimator, two things matter: that the reported accuracy holds for people the model has never seen, and that an estimate comes with an honest sense of its uncertainty. A single number with no range suggests more precision than a dataset of about 1,300 people can support.

This project started as an audit of an existing implementation (611noorsaeed/Medical-Insurance-Cost-Predictor, a scikit-learn Random Forest behind a Streamlit form). It became a rebuild with three goals:

1. **Honest evaluation:** measure accuracy without leakage and choose models only on data the final test never touches.
2. **Better and fairer predictions:** improve the error without relying on sex or region.
3. **Useful output:** give a calibrated range, show which changes move the estimate, and warn when the input is outside the data.

2. Dataset

The data is the widely used US medical-cost dataset (Lantz, 2013). Each row is one beneficiary: *age*, *sex*, *BMI*, number of *children* covered, *smoker* status, US *region*, and annual medical *charges* billed to the insurer (the target). After removing exact duplicates, 1,337 unique rows remain. Ages run from 18 to 64, and 20% of people smoke. Charges are strongly right-skewed: the mean is $13,279 but the median is only $9,386.

Figure 1. Annual charges by age. Charges rise steadily with age in three separate bands: non-smokers at the bottom, smokers in the upper two.

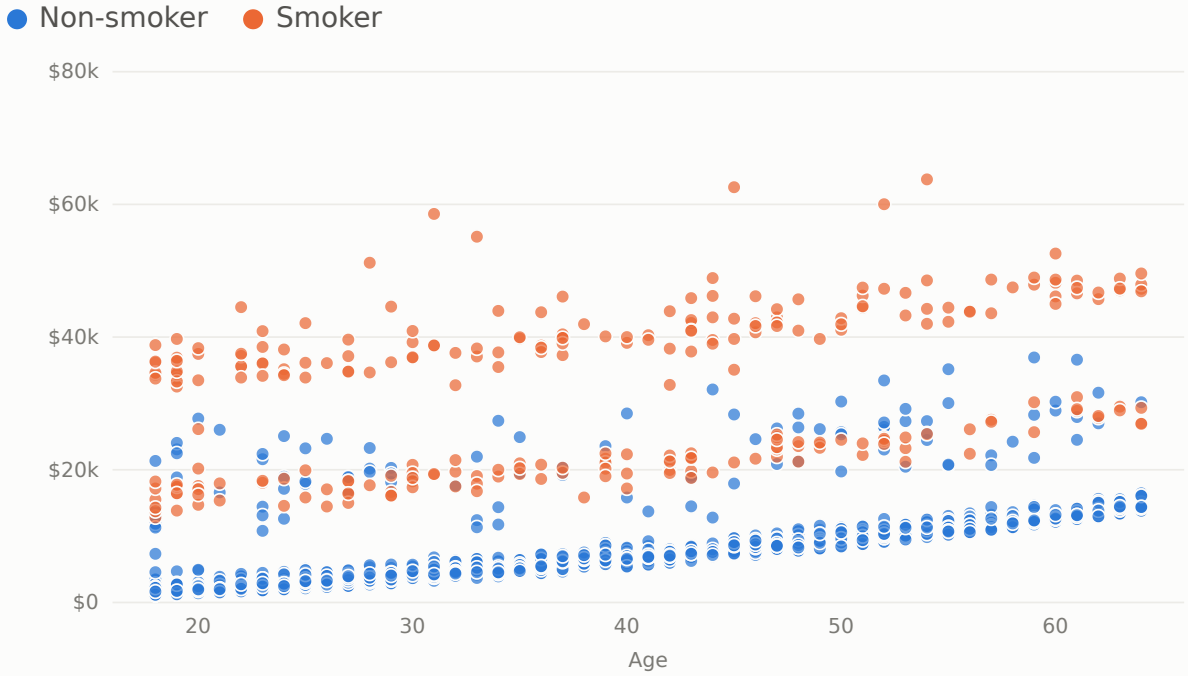
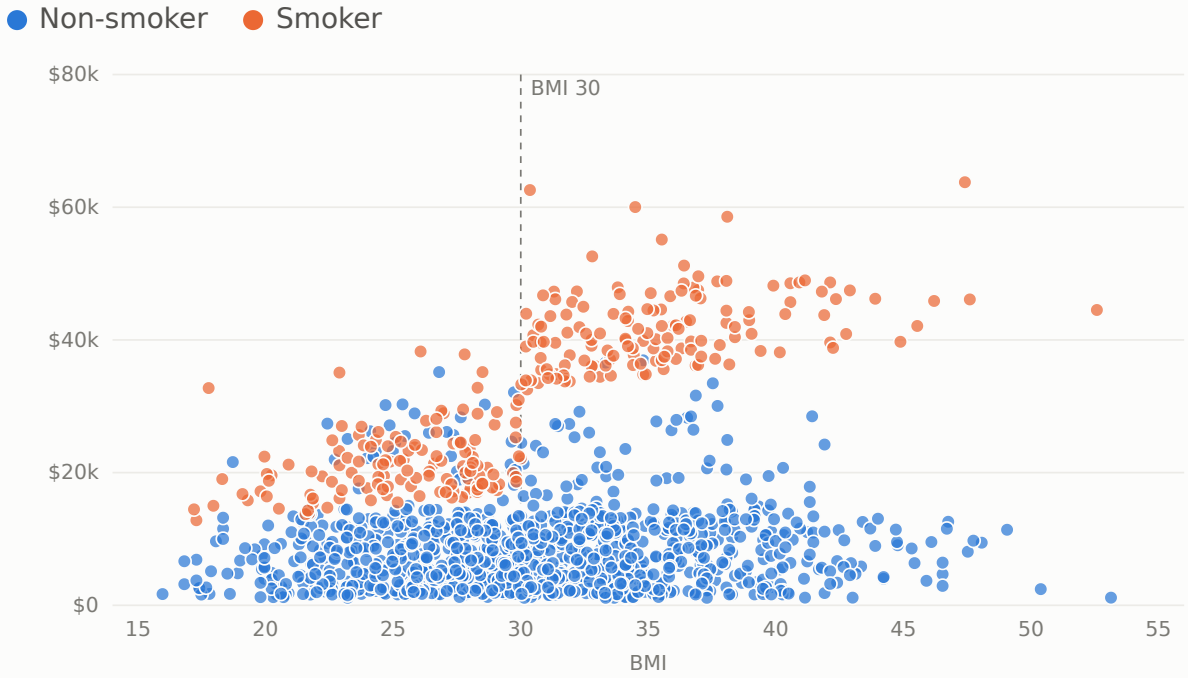


Figure 1 shows the main structure. Charges rise almost linearly with age, but people fall into separate bands, and smoking status decides which band. Figure 2 explains the split between the two smoker bands: among smokers, charges jump at BMI 30, the clinical threshold for obesity. Among non-smokers, BMI has almost no effect.

Figure 2. Annual charges by BMI. For smokers, charges step up at BMI 30 (dashed line). For non-smokers they stay flat.



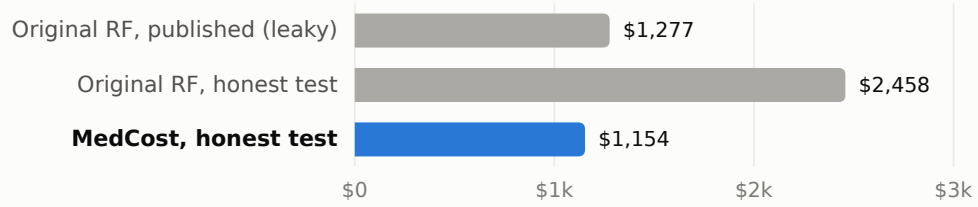
Group	People	Mean charges	Median charges
Non-smokers	1063	$8,441	$7,346
Smokers, BMI < 30	129	$21,363	$20,167
Smokers, BMI ≥ 30	145	$41,558	$40,904

Table 1. Charges by group. Obese smokers pay about twice as much as other smokers and about five times as much as non-smokers.

3. Auditing the original project

The original notebook loads a CSV of 2,772 rows, applies an 80/20 random split and reports $R^2 \approx 0.95$. However, the CSV is the original dataset appended to itself, so only 1,337 rows are unique. We reproduced the protocol exactly, with the same encoding, split seed and model. 445 of the 555 test rows (80%) have an identical copy in the training set, including the target. A Random Forest can memorise those rows, so the published score measures recall of seen data rather than generalisation. This is a textbook case of train/test contamination (Kaufman et al., 2012).

Figure 3. The same Random Forest scored three ways (mean absolute error, lower is better). Removing the leak roughly doubles the original model's error. MedCost is scored on the same honest test set.



The original app has two smaller problems. It prices by sex and region, and it accepts any age from 0 to 100, although the data only covers 18–64, without warning the user.

4. Method

4.1 Evaluation protocol

We deduplicate first, then hold out a stratified 20% test set of 268 people (stratified on smoker, since smokers are only a fifth of the data but dominate the error). We don't look at this test set until the model is chosen. All model comparison uses 5-fold cross-validation repeated 3 times (15 fits per candidate) on the remaining 80%. We report the mean and standard deviation across folds.

4.2 Features

Figures 1–2 suggest the interaction should be stated explicitly rather than left for the model to find. Beyond the raw inputs, the feature set adds `age²`, an `obese` flag (BMI ≥ 30), `smoker × obese` and `smoker × BMI`. Categoricals are one-hot encoded, unlike the original's arbitrary ordinal region codes. The same `engineer()` function is imported by both training and serving, so the two cannot drift apart.

4.3 Candidate models

We compared eight candidates: a mean-prediction baseline; the original linear and Random Forest models on their original encoding; a linear model, a tuned Random Forest and gradient boosting (Friedman, 2001) on the new features; gradient boosting with a log-transformed target; and gradient boosting with the Huber loss (Huber, 1964). The Huber loss is squared for small residuals and linear for large ones, so a few non-smokers with unexplained high charges (visible above the bottom band in Figure 1) do not pull the fit. All models use scikit-learn (Pedregosa et al., 2011).

4.4 Selection rule and fairness

The candidate with the lowest cross-validated MAE wins. Each candidate on the new features was also trained without sex and region. If that version is within one standard deviation of the winner's MAE, we ship it. The reasoning: these attributes should not set an individual's price unless they add real, measurable accuracy. In practice they add almost none.

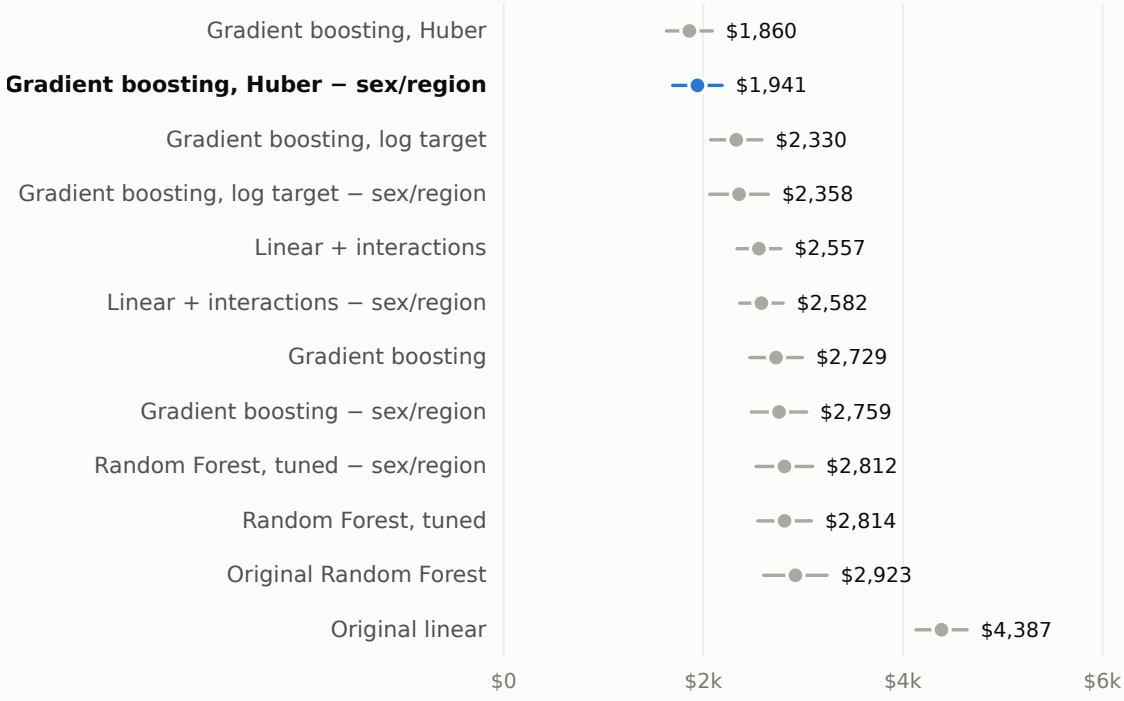
4.5 Prediction ranges

Each estimate comes with a 90% range built by split conformal prediction (Vovk et al., 2005; Lei et al., 2018). Out-of-fold residuals on the training data give empirical lower and upper quantiles (with the finite-sample correction), which are added to the point prediction. Residuals for smokers and non-smokers differ in size and shape, so the two groups are calibrated separately. For deployment, the model is refitted on all unique rows and the offsets are recalibrated.

5. Results

5.1 Model comparison

Figure 4. Cross-validated MAE for every candidate (dot = mean over 15 folds, whisker = ±1 standard deviation). The shipped model is highlighted. "− sex/region" marks the version trained without those two inputs.



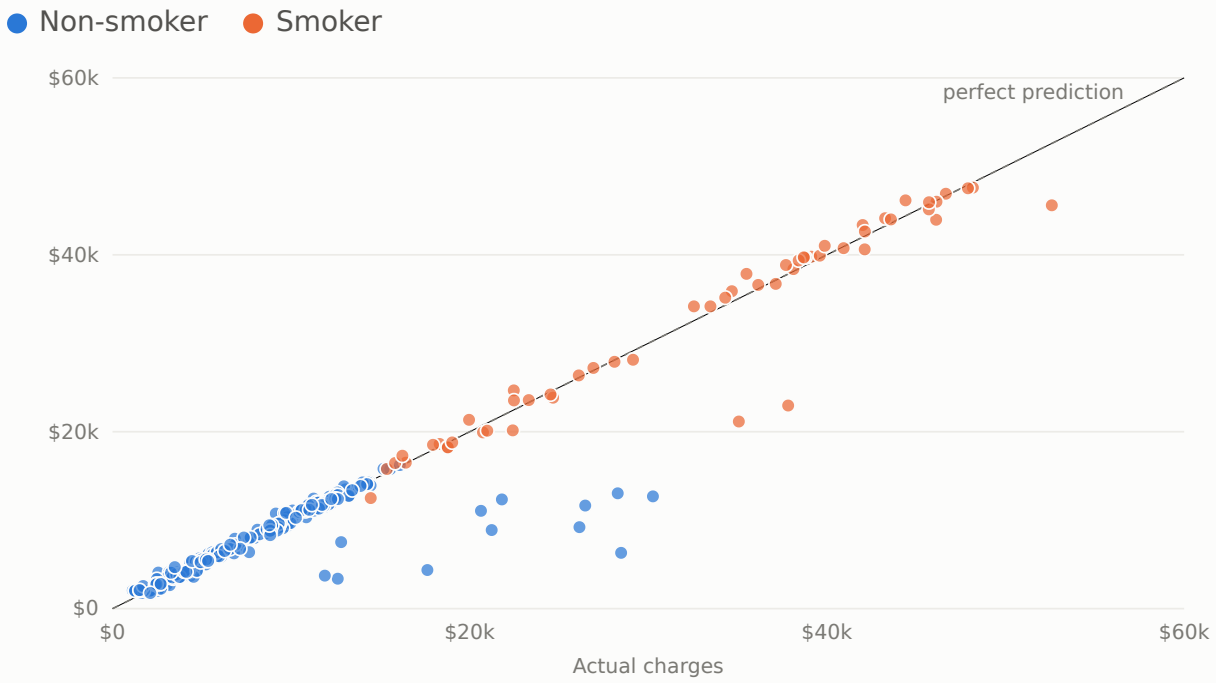
Feature engineering does most of the work. The same linear model drops from $4,387 to $2,557 MAE once it gets the smoker × obesity terms, and then beats the original Random Forest ($2,923). The Huber loss gives the largest single gain ($1,860 against $2,729 for squared loss). Its RMSE stays close to the other boosting models, which fits the explanation: it fits the typical case more tightly and gives up nothing it could have predicted on the outliers. Dropping sex and region costs $81 of MAE, well inside one fold standard deviation ($230), so the fair version ships.

5.2 Held-out test

Model / protocol	R^2	MAE	RMSE
Original RF, published protocol (leaky)	0.953	$1,277	$2,682
Original RF, honest test	0.884	$2,458	$4,093
MedCost, honest test	**0.928**	**$1,154**	**$3,228**

Table 2. The untouched test set (268 unique people, never used for any choice).

Figure 5. Predicted against actual charges on the held-out test set. Points on the dashed diagonal are exact. Points far below it are people charged much more than their inputs predict.

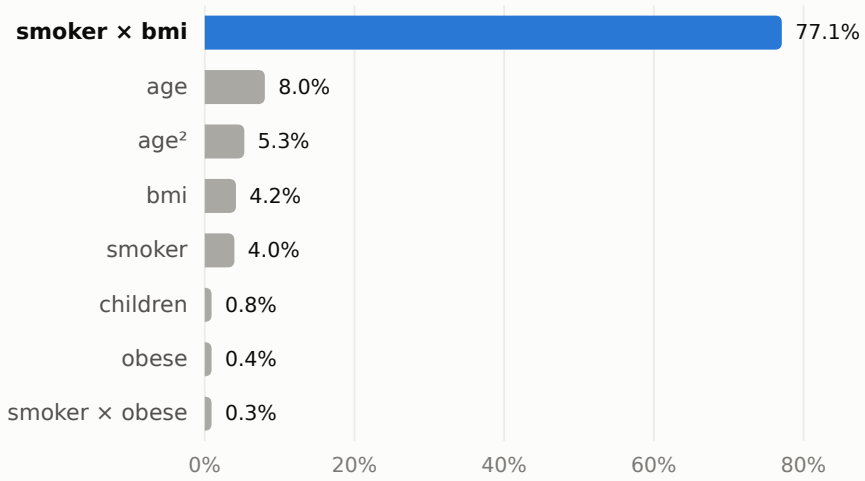


On the held-out test set, MedCost halves the original model's error: $1,154 against $2,458 MAE. Its honest R^2 (0.928) comes within 0.03 of the *leaky* number the original project published. Most points in Figure 5 sit close to the diagonal. The remaining error

is concentrated in a small set of non-smokers with high charges that none of the six inputs explain. More modelling would not fix that. Better data would.

5.3 What drives the estimate

Figure 6. Impurity-based feature importance of the shipped model (share of total loss reduction).



`smoker × BMI` alone accounts for 77% of the model's loss reduction. Age terms come second. Children barely matter.

5.4 Are the ranges honest?

Group	Test people	Inside 90% range
Smokers	55	91%
Non-smokers	213	96%
All	**268**	**95%**

Table 3. Share of held-out people whose true charge fell inside their 90% range.

Coverage meets or exceeds the 90% target in both groups, so the ranges are, if anything, slightly conservative. That is expected: they were calibrated on out-of-fold models trained on 80% of the training data, which are a little less accurate than the final model. The ranges are also asymmetric. For non-smokers the upper offset (+$14,699) is much wider than the lower one ($-899). The data shows the same thing: a non-smoker can land far above the typical charge but rarely far below it.

6. System

The model is served by a FastAPI application with a single static front end, with no framework and no build step. `POST /api/predict` returns the estimate, its 90% range, warnings for inputs outside the training data, what-if scenarios (quit smoking, BMI 25, both, ten years older) and an age curve for the user's profile and its smoking counterpart. `GET /api/metrics` exposes the full evaluation that this report and the site's "How accurate?" section are drawn from. The trained bundle (about 110 kB) is baked into a Docker image and runs as a non-root user behind Traefik with Let's Encrypt TLS. A development instance with live reload and Basic auth runs alongside it. A test suite checks predictions, input validation, invariance to sex and region, and the presence of the leakage finding.

7. Limitations and ethics

- **Small, old, US-only data.** Only 1,337 people, from an unspecified year and insurer. Charges are what was billed, not premiums, so the estimates are not quotes.
- **Missing drivers.** Pre-existing conditions, plan type and income are not recorded. The high-charge non-smokers in Figure 5 are most likely explained by exactly these.
- **Extrapolation.** Ages outside 18–64 and BMIs outside the observed range get a warning, but the model still returns a number.
- **Fairness is narrow.** Removing sex and region shows the model does not price on them directly. Other features could still act as proxies, and smoking and BMI themselves raise their own fairness questions.
- **Marginal coverage.** The 90% ranges hold on average within each smoker group. They are not guaranteed for every individual profile.

8. Conclusion

The original project's headline accuracy came from duplicated rows, not from learning. With leakage removed, one interaction feature and a loss that handles outliers, a slightly more careful pipeline halves the honest error. It also stops using sex and region and attaches a range that held for 95% of unseen people. Future work: conditional (per-profile) intervals such as conformalised quantile regression, and a recent, larger dataset with pre-existing-condition information.

References

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